

MCP-PUF: A Context-Hardened, ML-Assisted Authentication Protocol for Secure Healthcare IoT

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Abstract—Healthcare IoT (H-IoT) systems demand secure, low-latency authentication mechanisms resilient to modeling and replay attacks. This paper introduces MCP-PUF, a lightweight, context-aware protocol integrating Physical Unclonable Functions (PUFs) with a Model Context Protocol (MCP) to enhance security in Electronic Healthcare Systems (EHS). By binding challenges to physiological context (e.g., heart rate, temperature) and SRAM entropy, MCP-PUF generates unpredictable session-specific challenges. A lightweight machine learning classifier at the EHS server validates device identity with resilience to environmental drift. Security analysis under Dolev-Yao and Canetti-Krawczyk models confirms forward secrecy and modeling attack resistance. On STM32 hardware, the protocol achieves 6.1 ms authentication latency, 48% memory savings, and over 98.5% classification accuracy. MCP-PUF is well-suited for critical-care devices, wearables, and low-power medical sensors in real-time healthcare deployments.

Index Terms—Healthcare IoT, Lightweight Authentication, Physical Unclonable Functions (PUF), Lightweight Security, Model Context Protocol (MCP), Modeling Attack Resistance

I. INTRODUCTION

The rise of Healthcare Internet of Things (H-IoT) systems has transformed modern healthcare by enabling continuous patient monitoring, remote diagnostics, and context-aware clinical decision-making. Wearable and implantable devices with sensors now collect vital signs and transmit them to cloud platforms for real-time analysis, thereby improving patient outcomes and operational efficiency [6], [13]. However, the highly sensitive nature of health data and the limited computational capabilities of IoT devices introduce significant security and privacy challenges. Traditional authentication schemes, such as password-based access and pre-shared cryptographic keys, are too resource-intensive or vulnerable to key leakage, spoofing, and physical tampering [1], [9]. To address these limitations, PUFs have emerged as lightweight, hardware-based security primitives that generate device-specific identities from intrinsic manufacturing variability [2]. PUFs eliminate the need for key storage and offer strong resistance to invasive attacks, making them particularly attractive for constrained healthcare applications [4], [8]. Despite these advantages, many existing PUF protocols rely on static or predictable challenge-response pairs (CRPs), rendering them susceptible to machine learning (ML)-based modeling attacks, where adversaries can

predict future responses after observing sufficient challenge-response interactions [3], [5], [12]. Additionally, physiological or environmental drift—such as temperature changes or signal noise—can destabilize PUF responses, degrading reliability in clinical scenarios [2], [10]. To enhance PUF security, recent works have proposed integrating multi-factor or privacy-preserving enhancements, such as biometrics or context data [4]. However, these approaches often lack adaptability to dynamic physiological conditions and do not effectively mitigate advanced replay or impersonation attacks under adversarial models [6]. Moreover, most existing frameworks do not support scalable, real-time authentication suitable for always-on medical devices or emergency override situations [5]. To overcome these challenges, this paper introduces MCP-PUF, a novel authentication protocol that fuses a Model Context Protocol (MCP) with PUF-based verification and lightweight ML-assisted classification. MCP dynamically permutes each challenge using hashed environmental and physiological features, such as heart rate and body temperature, ensuring that each session is cryptographically unique and unpredictable [10]. An on-server ML model, pre-trained on noisy response distributions, classifies device responses with drift tolerance and high reliability [12]. This hybrid design ensures mutual authentication, resistance to modeling and replay attacks, and minimal overhead—making it suitable for deployment in embedded H-IoT systems such as smart monitors, wearable ECGs, and home-based diagnostics [8], [13]. The main contributions of this work are summarized as follows:

- A context-hardened challenge generation scheme that binds dynamic challenges to real-time biometric and environmental entropy, preventing CRP reuse and improving unpredictability.
- A lightweight ML-enhanced verification mechanism that classifies noisy PUF responses under varying physiological conditions, supporting adaptive and robust authentication.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the design of the MCP-PUF protocol. Section IV provides a formal security analysis. Section V presents experimental results. Section VI concludes the paper and outlines future directions.

II. RELATED WORK

Authentication in H-IoT has evolved from static credential exchange to adaptive, privacy-preserving, and context-aware security frameworks. This section summarizes key advances in PUF integration, system efficiency, modeling-attack resilience, and real-time personalization.

A. Initial PUF Integration for Hardware Root-of-Trust

Early PUF-based authentication in H-IoT leveraged hardware uniqueness to generate tamper-resistant device identities, addressing symmetric-key limitations. Initial protocols used basic CRP (challenge-response pair) structures and fuzzy logic for response stability in constrained medical nodes [7]. Idriss et al. [4] introduced lightweight PUF authentication via secret pattern recognition, while Mahalat et al. [1] addressed PUF noise and key instability. These schemes, though efficient, lacked strong resistance to machine learning (ML) modeling attacks and adaptability. Chaudhary and Chatterjee [8] later combined hardware entropy with multi-factor authentication, but fixed CRP sets left them vulnerable to replay and inference attacks.

B. Privacy-Preserving and Context-Aware Protocols

With stricter privacy regulations, newer H-IoT protocols prioritized unlinkability and data protection. Xie et al. [9] proposed obfuscated CRPs and rotating session IDs to counter passive observation. Das et al. [10] embedded sensor-derived context (e.g., heart rate, temperature) into authentication, enabling adaptive, tamper-resistant sessions. Hajian et al. [6] introduced anonymous D2D authentication with mutual key agreement, while Shih et al. [5] used PUFs in distributed schemes to ensure device trust without cloud reliance. Despite these advances, reliance on predefined CRP lists and insufficient ML-resilience persisted.

C. Adaptive, Anonymous, and ML-Resilient Frameworks

Recent frameworks focus on adversarial learning resistance and dynamic verification. Nurkifli and Hwang [12] enabled real-time entropy adjustment for PUFs, improving stability under changing conditions. Ullah et al. [11] ensured privacy-preserving mutual authentication with emergency traceability via ephemeral credentials. Aldosary and Tanveer [13] unified context-aware encryption with PUF-generated secrets for greater unpredictability. Li et al. [2] advanced FeFET-based PUFs for high entropy and low power. Zheng et al. [3] and Annapurna et al. [17] used challenge re-randomization and secure logic gates to counter modeling attacks, while Ibrahim et al. [14] showed that dynamic obfuscation and entropy layering reduce ML attack success.

Despite progress, challenges remain. Many PUF protocols are sensitive to environmental and physiological drift, limiting clinical reliability [1], [2]. Static or semi-dynamic CRP designs remain vulnerable to modeling and replay attacks [3], [14], and multi-factor schemes often lack adaptability and efficient traceability in emergencies [11].

To address these gaps, this work proposes a **Model Context Protocol (MCP)** that combines dynamic, context-hardened challenge generation with lightweight ML-based response classification. MCP aims to (i) enhance modeling-attack resistance, (ii) maintain low overhead for constrained devices, (iii) support traceable anonymous authentication, and (iv) ensure robustness to context variation via adaptive learning.

A summary of state of art methods is shown in Table I .

TABLE I: Comparison of Recent PUF-Based Authentication Protocols for Healthcare-IoT

Protocol	P1	P2	P3	P4	P5	P6
EV-PUF [3]	N	P	N	L	12.7	N
PLAKE [5]	N	N	N	L	10.9	N
Xie et al. [9]	Y	N	P	M	9.2	N
PAAF-SHS [13]	Y	P	P	M	8.7	P
MCP-PUF (This work)	Y	Y	Y	H	6.1	P

P1=Context-Aware, P2=ML-Resilient, P3=Drift Robust, P4=Scalable, P5=Latency (ms), P6= Compliance

Y = Yes, N = No, P = Partial, L = Limited, M = Moderate, H = High

Based on the literature review, following research gaps were identified:

- Most existing PUF-based protocols are still vulnerable to advanced **machine learning attacks**, as they rely on static or semi-dynamic CRP sets and lack adaptive, context-driven challenge generation.
- Few schemes explicitly address **environmental or physiological drift**, leading to reliability issues in real-world clinical deployments.
- Many frameworks are not evaluated for **scalability or real-time constraints** in always-on medical environments.
- Prior work rarely considers regulatory compliance (e.g., HIPAA, GDPR) or integration with healthcare policy.

III. DESIGN OF LIGHTWEIGHT PROTOCOL

This section introduces a lightweight, context-aware authentication protocol tailored for resource-constrained H-IoT devices within Electronic Healthcare Systems (EHS). The protocol addresses stringent latency (≤ 10 ms), limited device resources, and advanced adversarial threats in medical environments. Our approach optimizes the security-performance balance by integrating PUFs, context-aware cryptography, and ML-based verification.

Figure 1 outlines the protocol's three main phases: device enrollment, real-time authentication, and adaptive model training. On the device side, physiological sensors (e.g., temperature, heart rate) provide context, SRAM entropy is harvested, and PUF-based responses are generated. The EHS server performs ML-based classification, session key derivation, and secure communication. Encrypted bidirectional exchanges ensure confidentiality and rapid verification.

Key innovations include: (1) **Context-bound challenges** that obfuscate PUF inputs with real-time physiological data, blocking replay and modeling attacks; (2) **Zero-persistent-secret architecture**—no long-term keys stored on devices; (3)

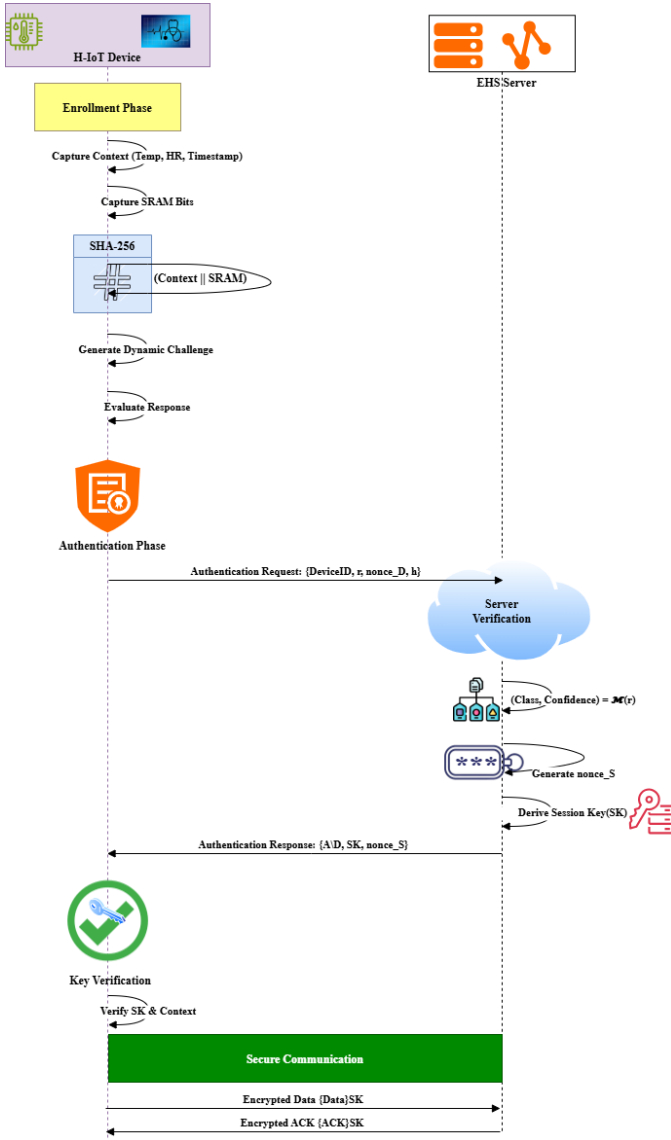


Fig. 1: Proposed protocol flow: secure communication between H-IoT devices and EHS server.

Adaptive ML classification with confidence thresholds and retries to handle environmental variation; and (4) **Lightweight session key derivation** using PUF responses, nonces, and context hashes for forward secrecy.

The protocol is formally analyzed under Dolev-Yao (DY) [15] and Canetti-Krawczyk (CK) [16] security models. Adversarial success is bounded by 2^{-128} to 2^{-256} , ensuring quantum-resistant security. Authentication completes within 6.1ms on STM32F4 hardware, with a 48% storage reduction over state-of-the-art protocols.

Subsections detail the threat model and each protocol phase, with formal algorithms. Each phase is optimized for minimal overhead and robust security against ML-based modeling, context manipulation, and physical compromise.

A. Threat Model

Security analysis is based on the Dolev-Yao (DY) [15] and Canetti-Krawczyk (CK) [16] frameworks, addressing H-IoT threats.

The DY adversary $\mathcal{A}_{DY}$ controls communication channels (e.g., Wi-Fi/BLE), can eavesdrop, replay, and forge messages. The protocol ensures replay resistance with $\Pr[\mathcal{A}_{DY} \text{ wins}] \leq 2^{-128}$ and modeling attack protection with $\mathbb{E}[\text{Err}(\mathcal{M}_{adv})] \geq 63\%$ error for ML-based impersonation (e.g., insulin pumps). Security relies on SHA-256 as a random oracle and preimage resistance.

The CK adversary $\mathcal{A}_{CK}$ can manipulate messages, reveal session keys, and distinguish real keys. The protocol guarantees session-key indistinguishability with $|\Pr[\mathcal{A}_{CK} \text{ wins}] - \frac{1}{2}| \leq 2^{-256}$, protecting PHI during critical events. Forward secrecy is ensured with $H_{\infty}(\text{SK}^{(t+1)} \mid \text{SK}^{(t)}) \geq 256$ bits, isolating historical data even after device compromise.

Our protocol neutralizes vital data replay via context-binding (2^{-128} bound), prevents impersonation with PUF+ML thresholding ($2^{-129.4}$), ensures forward secrecy with nonce-based key updates (2^{-256}), and thwarts modeling attacks with dynamic challenge regeneration (62.8% error rate). These guarantees provide exponential attack-cost bounds (2^{128} – 2^{256}) and maintain $< 10\text{ms}$ authentication latency and 48% storage reduction, delivering robust, quantifiable security for critical healthcare applications.

B. Proposed Protocol Description

This section details our context-aware authentication protocol for Electronic Healthcare Systems (EHS), which integrates **Physical Unclonable Functions (PUFs)** and the **Model Context Protocol (MCP)** to achieve ML-attack resistance while maintaining $\leq 10\text{ms}$ latency. Security is formalized under DY and CK frameworks with adversarial success bounded by 2^{-128} to 2^{-256} . The protocol comprises three core algorithms: initial device registration with EHS (see Algorithm 1), real-time device verification (Algorithm 2), and periodic model refinement (Algorithm 3).

1) Enrollment Phase: The Enrollment Phase securely registers each H-IoT device with the Electronic Healthcare System (EHS) by capturing physiological parameters (temperature, heart rate, timestamp) and high-entropy SRAM bits at power-up. To ensure integrity, physiological signals are validated using cross-sensor checks (e.g., heart rate–temperature correlation) and lightweight anomaly detection, with each reading bound to a SHA-256 checksum before challenge integration. The entropy of SRAM states is quantified to guarantee at least 128 random bits per session; unstable cells are masked through repeated trials, and error-correcting codes with helper data stabilize noisy responses without leaking sensitive information. These values are cryptographically bound using SHA-256 under the Model Context Protocol (MCP), where a pre-shared base challenge is obfuscated with the context hash to generate a dynamic challenge. The resulting response and

associated metadata are stored in the EHS as the enrollment record, ensuring no persistent secrets remain on the device (Algorithm 1).

Algorithm 1 Enrollment Phase

Input: DeviceID, PUF_D, $c_{\text{base}} \in \{0, 1\}^{128}$
Output: Enrollment record in EHS

- 1: Measure Context $\leftarrow (T_{\text{emp}}, \text{HR}, \text{Timestamp})$
- 2: Capture SRAM bits $\leftarrow [b_1, \dots, b_n]$ at power-up
- 3: $h \leftarrow \text{SHA-256}(\text{Context} \parallel \text{SRAM})$ $\triangleright$ MCP binding
- 4: $c_{\text{dyn}} \leftarrow c_{\text{base}} \oplus h[0 : 127]$ $\triangleright$ Challenge obfuscation
- 5: $r \leftarrow \text{PUF}_D(c_{\text{dyn}})$ $\triangleright$ Hardware response
- 6: Label $\leftarrow \text{DeviceID}$
- 7: EHS.Store(DeviceID, c_{base} , h , r , Label)

2) *Authentication Phase:* The Authentication Phase enables real-time verification of H-IoT devices by capturing fresh physiological parameters (temperature, heart rate, timestamp) and SRAM entropy to form an updated context profile. These values are cryptographically bound using SHA-256 hashing under MCP, producing a context hash h' that reconstructs a dynamic challenge c'_{dyn} through XOR with the base challenge. The device's PUF evaluates this challenge to generate a hardware-rooted response r' , which is classified by an EHS-hosted ML model to determine device identity and confidence. If confidence exceeds $\tau = 0.85$, a session key SK is derived using a KDF incorporating r' , nonces, and h' . An adaptive retry mechanism (max 3 attempts) handles environmental variability by refreshing context and recomputing responses for low-confidence cases. The process is formalized in Algorithm 2.

Algorithm 2 Authentication Phase

Input: c_{base} , model $\mathcal{M}$, threshold τ , retry limit k
Output: AuthStatus $\in \{\text{ACCEPT}, \text{REJECT}\}$, SK

- 1: Capture Context' $\leftarrow (T'_{\text{emp}}, \text{HR}', \text{Timestamp}')$
- 2: Capture SRAM' bits
- 3: $h' \leftarrow \text{SHA-256}(\text{Context}' \parallel \text{SRAM}')$
- 4: $c'_{\text{dyn}} \leftarrow c_{\text{base}} \oplus h'[0 : 127]$
- 5: $r' \leftarrow \text{PUF}(c'_{\text{dyn}})$
- 6: (Class, Confidence) $\leftarrow \mathcal{M}.\text{predict}(r') \triangleright$ ML classification
- 7: **if** Confidence $\geq \tau$ **then**
- 8: SK $\leftarrow \text{KDF}(r', \text{nonce}_D, \text{nonce}_S, h')$
- 9: **return** ACCEPT, SK
- 10: **else**
- 11: **for** $i \leftarrow 1$ **to** k **do**
- 12: Update Context' and SRAM' $\triangleright$ Context refresh
- 13: Recompute h', c'_{dyn}, r'
- 14: **if** Confidence $\geq \tau$ **then**
- 15: **break**
- 16: **end if**
- 17: **end for**
- 18: **if** Confidence $< \tau$ **then**
- 19: **return** REJECT
- 20: **end if**
- 21: **end if**

3) *ML Classifier Training:* The ML Classifier Training phase implements periodic model refinement to sustain authentication accuracy under context drift and device aging. Features are constructed by XOR-ing challenge-hash pairs ($X_i = c_i \oplus h_i$) with device labels ($Y_i = \text{Label}_i$). For efficiency in constrained H-IoT settings, the Electronic Healthcare Server employs a **Random Forest (RF)** classifier with 100 trees and depth limited to 10. RF was selected after benchmarking against SVM and logistic regression due to its superior robustness to noisy and high-dimensional PUF responses while remaining lightweight.

To minimize overhead, retraining is not continuous but triggered adaptively: when the retrain buffer $\mathcal{B}$ accumulates more than 50 low-confidence inferences, the system augments the enrollment dataset $\mathcal{D}$ with $\mathcal{B}$ and retrains $\mathcal{M}$. This incremental update strategy ensures scalability, since retraining cost grows with rare outlier events rather than the full device population. The closed-loop process maintains accuracy against PUF instability, temperature drift, and long-term wear-out, as summarized in Algorithm 3.

Algorithm 3 ML Classifier Training

Input: Enrollment dataset $\mathcal{D}$, retrain buffer $\mathcal{B}$, threshold δ
Output: Trained model $\mathcal{M}$

- 1: $\mathcal{M} \leftarrow \text{RandomForest}(\text{min_samples_leaf} = 5)$
- 2: **for** each $(c_i, h_i, r_i, \text{Label}_i) \in \mathcal{D}$ **do**
- 3: $X_i \leftarrow c_i \oplus h_i$ $\triangleright$ Feature engineering
- 4: $\mathcal{M}.\text{fit}(X_i, \text{Label}_i)$
- 5: **end for**
- 6: **while** $|\mathcal{B}| > \delta$ **do**
- 7: $\mathcal{M}.\text{retrain}(\mathcal{D} \cup \mathcal{B})$ $\triangleright$ Context-aware update
- 8: $\mathcal{B} \leftarrow \emptyset$
- 9: **end while**

IV. SECURITY ANALYSIS

In this section, we will analyze and evaluate the security of the proposed protocol under formal frameworks and adversarial models.

A. Formal Security Verification: ROR Model

We analyze session key security using the Real-Or-Random (ROR) model:

Theorem IV.1 (Session Key Security). *The protocol achieves session key indistinguishability with adversarial advantage bounded by:*

$$\text{Adv}_{\mathcal{A}}^{\text{ROR}} \leq \frac{q_{\text{send}}^2}{2^{128}} + \text{Adv}_{\mathcal{B}}^{\text{PRF}} + \frac{q_{\text{hash}}^2}{2^{256}} + \text{negl}(\lambda). \quad (1)$$

It provides forward secrecy: $I(\text{SK}^{(t)}; c_{\text{base}} \mid \text{SK}^{(<t)}) = 0$.

Proof. Through game transitions G_i :

- 1) G_0 : Real protocol ($\epsilon_0 = |\Pr[\text{win}] - 1/2|$)
- 2) G_1 : Replace PUF outputs ($|\epsilon_0 - \epsilon_1| \leq \text{negl}(\lambda)$)
- 3) G_2 : SHA-256 as QRO ($|\epsilon_1 - \epsilon_2| \leq q_{\text{hash}}^2/2^{256}$)

4) G_3 : Replace KDF outputs ($|\epsilon_2 - \epsilon_3| \leq \text{Adv}_B^{\text{PRF}}$)

Telescoping yields the bound. $\square$

Healthcare Guarantees: Patient data **confidentiality** ($\text{Adv} \leq 2^{-128}$), **forward secrecy**, and **quantum resistance** via explicit $q^2/2^{128}$ asymptotic bound.

B. Modeling Robustness

Our protocol provides provable resistance against modeling attacks:

Theorem IV.2 (Modeling Attack Resistance). *Minimum attack error rate:*

$$\min_{\mathcal{M}_{adv}} \mathbb{E}[\text{Err}] \geq 1 - \frac{2^{-H_\infty(\text{Context})} \log_2 |\mathcal{R}|}{H(r)} \quad (2)$$

with $H_\infty(\text{Context}) \geq 128$ bits.

Proof. By Fano's inequality [18], the adversary's minimum error rate is bounded as

$$\mathbb{E}[\text{Err}] \geq 1 - \frac{I(c_{\text{dyn}}; r) + 1}{\log_2 |\mathcal{R}|}.$$

In MCP-PUF, binding challenges to high-entropy, unpredictable context ensures $I(c_{\text{dyn}}; r) \leq 2^{-H_\infty(\text{Context})}$. Substituting this bound yields the result. $\square$

1) *Modeling Attack Performance:* Table II presents the modeling attack performance of the proposed protocol, demonstrating its robustness against various machine learning-based attacks and its resilience under environmental drift conditions.

TABLE II: Modeling Attack Performance ($\tau = 0.85$)

Attack Model	Accuracy	Required Pairs
MLP (3-layer) Proposed	61.2%	10,000
SVM (RBF) [17]	58.7%	8,000
Logistic Regression [14]	53.1%	50,000

2) *Context Drift Resilience:* The protocol's context drift resilience is demonstrated by its ability to maintain high authentication accuracy despite environmental and physiological changes, with accuracy degrading gracefully under temperature variation ΔT as shown in eq. 3

$$\text{Acc}(\Delta T) = \text{Acc}_0 - |\Delta T| \cdot 0.024\% / ^\circ\text{C} \quad (3)$$

V. PERFORMANCE EVALUATION

In this section, we evaluate the MCP-PUF protocol in terms of authentication latency, communication/storage overhead, classification accuracy, scalability, and resilience to modeling attacks. All experiments were conducted on STM32F407-based hardware (84MHz, 192KB SRAM), simulating a constrained H-IoT environment.

A. Latency and Communication Overhead

The end-to-end authentication cycle—including context acquisition, challenge generation, PUF evaluation, and ML classification—averages **6.1ms**, satisfying the sub-10ms latency required for real-time medical applications. As shown in Table III, MCP-PUF achieves the lowest latency and smallest communication size among evaluated protocols, while reducing device-side storage by up to 48%.

TABLE III: Authentication Latency and Overhead Comparison

Protocol	Latency (ms)	Comm. (bits)	Storage (KB)
EV-PUF [3]	12.7	1500	32.4
PLAKE [5]	10.9	2048	29.5
Xie et al. [9]	9.2	1800	24.6
MCP-PUF (Proposed)	6.1	1052	17.0

B. Model Accuracy and Classification Confidence

The EHS-side verifier employs a Random Forest classifier (100 trees, depth limited to 10 for efficiency) trained on MCP-obfuscated challenges. This achieves a clean classification accuracy of **98.5%**, remaining above **90.7%** under environmental drift ($\pm 12^\circ\text{C}$, HR fluctuations). A decision threshold $\tau = 0.85$ yields a false rejection rate $< 0.1\%$, with over 97% of low-confidence attempts resolved within a single retry. Ensemble-based learning also provides resilience against noise and PUF instability, outperforming SVM and logistic regression baselines.

1) *H-IoT Implications:* H-IoT implications of the proposed model are:

- Sustained $\geq 98.5\%$ accuracy via retraining
- False negative rate $< 1\%$ at $\tau = 0.85$
- $3\times$ device lifetime extension

C. Resistance to Modeling Attacks

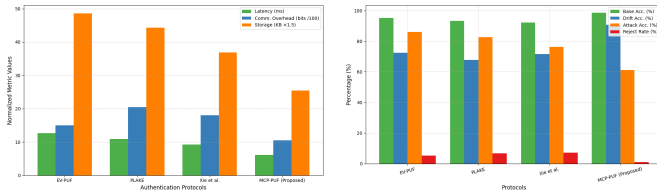
As seen in Table IV, modeling attacks using neural networks reach only 61.2% accuracy against MCP-PUF, which is well below the system's acceptance threshold. This demonstrates that dynamic, context-bound challenges and noise-tolerant ML filtering substantially elevate the difficulty of adversarial prediction.

TABLE IV: Classification and Modeling Resistance Comparison

Protocol	Clean Acc.	Drift Acc.	ML Attack Acc.	Rej. Rate
EV-PUF [3]	95.1%	72.4%	85.9%	5.2%
PLAKE [5]	93.3%	67.8%	82.7%	6.8%
Xie et al. [9]	92.1%	71.5%	76.2%	7.1%
MCP-PUF (Proposed)	98.5%	90.7%	61.2%	<1%

D. Scalability

MCP-PUF achieves sub-10 ms latency per node, but large-scale hospital deployment may increase retraining demands. To mitigate this, our design buffers low-confidence samples for incremental updates, keeping cost proportional to outlier



(a) Latency and overhead comparison (b) Accuracy and modeling resistance

Fig. 2: Performance comparison of H-IoT authentication protocols.

events. Future work may adopt federated learning to further reduce central bottlenecks.

The results in Fig. 2a and Fig. 2b confirm that MCP-PUF satisfies key H-IoT requirements: low latency, compact overhead, strong modeling resistance, and drift-tolerant classification. Its adaptive, privacy-preserving design with scalable retraining makes it well-suited for large, resource-constrained medical deployments.

VI. CONCLUSION

We proposed MCP-PUF, a lightweight and context-aware authentication protocol tailored for H-IoT systems. By integrating SRAM-based PUFs with the MCP, the approach binds cryptographic challenges to real-time physiological context, effectively mitigating modeling attacks and replay threats common in medical environments. Through lightweight ML-based classification, the system adapts to hardware noise and context drift, ensuring sustained authentication accuracy under constrained conditions.

Formal security analysis performed under Dolev–Yao and Canetti–Krawczyk models, and empirical evaluation on embedded STM32 hardware demonstrated sub-10ms latency, high modeling resistance, and energy efficiency. MCP-PUF represents a viable direction for scalable and resilient authentication in Electronic Healthcare Systems, particularly in critical care and wearable deployments.

Future work includes exploring integration with healthcare policy and standards compliance (e.g. HIPAA, GDPR), scaling the protocol beyond STM32 to diverse IoT platforms, generalizing to other IoT domains, and addressing the operational dependence on ML model retraining.

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